

TOPIC 12

Production Deployment, Monitoring & Version Control

Production-ready hardening package for the n8n AI Agent workflow

Main Workflow	LMS Topic 11 – Autonomous Support & Lead Generation Agent
Monitoring Workflow	Production AI Agent Monitoring
AI Model	Google Gemini 2.5 Flash
Repository	github.com/shaikshahid777/production-ai-agent
Prepared For	Orion LMS / Tayana Academy

Document status: Submission documentation package. Manual environment-dependent items are explicitly marked as pending rather than over-claimed.

1. Assessment Overview

This document summarizes the Topic 12 production deployment, monitoring, security hardening, and version-control implementation for the n8n AI Agent workflow. The package covers workflow export, environment-variable separation, security review, execution-log retention configuration, monitoring, alerting, and deployment readiness.

The assessment requires a production-oriented approach: separate configuration from workflow logic, remove hardcoded secrets, retain versioned workflow exports, monitor service health, detect failures, and alert operators automatically.

2. Solution Architecture

Main AI Agent: Webhook intake → required-field validation → Postgres context enrichment → Google Gemini 2.5 Flash AI support agent → structured output / decision routing → customer email + internal Slack notification + token-usage logging → webhook response.

Monitoring: Schedule Trigger (every 15 minutes) → HTTP health check → IF health decision → healthy/no-alert branch or Slack failure alert.

3. Production Hardening

- **Environment variables:** sender email and Slack channels are referenced through n8n \$env values instead of being hardcoded.
- **Monitoring configuration:** MONITOR_TARGET_URL and SLACK_ALERT_CHANNEL are environment-specific values.
- **Secret hygiene:** credentials are referenced by n8n credential IDs/names; secret values are not intended to be stored in Git.
- **AI provider:** Google Gemini 2.5 Flash is used; no OpenAI dependency is part of the production implementation.
- **Validation:** required webhook fields and structured AI output handling are included.
- **Failure handling:** validation-error and fallback paths are documented and included in the workflow design.

4. Environment Variables

Variable	Purpose
SUPPORT_FROM_EMAIL	Customer email sender in the main workflow
SLACK_SUPPORT_CHANNEL	Internal support notification channel
MONITOR_TARGET_URL	Health endpoint checked by the monitoring workflow
SLACK_ALERT_CHANNEL	Destination for monitoring failure alerts

5. Security Review

The repository is designed to exclude secrets and sensitive credential values. The security review identifies the following production controls and follow-up items.

Area	Status	Notes
Secrets in workflow JSON	PASS	No intended API keys, passwords, Slack tokens, or database passwords in workflow parameters.

Area	Status	Notes
Environment separation	PASS	\$env references used for environment-specific sender/channel/monitor settings.
Input validation	PASS	Required webhook fields are validated before downstream processing.
Structured AI output	PASS	Structured parsing and fallback handling are included.
Webhook protection	REVIEW	Use authentication/upstream gateway and rate limiting before exposing publicly.
PII minimization	REVIEW	Minimize customer data sent to Slack and retained in logs.
Least privilege	REVIEW	Use least-privilege database and service credentials.

6. Execution Log Retention

Execution-data pruning is an n8n instance/server-level configuration. The documented assessment configuration is:

```
EXECUTIONS_DATA_PRUNE=true
EXECUTIONS_DATA_MAX_AGE=336
EXECUTIONS_DATA_PRUNE_MAX_COUNT=10000
```

These values must be applied in the actual n8n host environment and followed by an n8n restart. Verification should confirm that old executions are removed and retention stays within the configured limits.

7. Monitoring Workflow

The monitoring workflow is named **Production AI Agent Monitoring**. It is designed to run every 15 minutes and probe the production agent health endpoint.

- **Schedule Trigger:** starts the health check every 15 minutes.
- **HTTP Request:** requests the endpoint configured through `MONITOR_TARGET_URL`.
- **IF node:** classifies 2xx/3xx responses as healthy.
- **Healthy branch:** terminates without an alert.
- **Unhealthy branch:** sends a Slack alert to `SLACK_ALERT_CHANNEL`.

8. Monitoring Validation

The workflow's unhealthy routing was verified during sandbox testing. The sandbox did not provide a live health endpoint and Slack delivery was simulated, so a real environment test is still required.

- **Healthy test:** live endpoint returns a successful response → IF evaluates true → no Slack alert.
- **Failure test:** use an invalid/broken health endpoint → IF evaluates false → Slack Alert executes.
- **Evidence:** capture the healthy execution, failed execution routing, and the actual Slack message.

9. Version Control & Repository

The public GitHub repository contains the workflow exports, documentation, screenshots placeholder, README, and .gitignore.

Repository: <https://github.com/shaikshahid777/production-ai-agent>

Recommended release practice: keep main as the production-ready branch, use descriptive commits, and tag known-good releases for rollback.

10. Repository Structure

```
production-ai-agent/  
  ■■■ workflows/  
    ■ ■■■ production-ai-agent.json  
    ■ ■■■ monitoring-workflow.json  
  ■■■ docs/  
  ■■■ screenshots/  
  ■■■ README.md  
  ■■■ .gitignore
```

11. Deployment & Rollback

- Configure the required environment variables in the n8n host.
- Connect Gemini, Postgres, SMTP/Gmail, and Slack credentials in n8n.
- Run DB setup against the real database where applicable.
- Configure and verify the live health endpoint.
- Apply execution pruning at the instance level and restart n8n.
- Run smoke tests and monitoring tests before production activation.
- For rollback, restore a known-good workflow export from Git/n8n version history and re-import/republish it.

12. Evidence & LMS Submission Checklist

Deliverable / Requirement	Status
Workflow exported	DONE
GitHub repository created and public	DONE
Main workflow JSON in workflows/	DONE
Monitoring workflow JSON in workflows/	DONE
Environment-variable references	DONE
Hardcoded secrets removed	DONE
Production configuration documented	DONE
Execution-log pruning applied on host	PENDING — manual
Monitoring workflow created	DONE
Schedule Trigger configured	DONE
HTTP Request configured	DONE
IF node configured	DONE
Slack Alert configured	DONE — delivery needs real test
Real monitoring failure + Slack test	PENDING — manual
Workflow screenshots	PENDING
Loom/YouTube demo	PENDING
Final LMS submission	PENDING

13. Demo Video Outline

- Introduce Topic 12 and the production AI Agent.
- Show the main workflow architecture and Gemini 2.5 Flash.
- Show \$env-based configuration and credential separation.
- Demonstrate the monitoring workflow: Schedule → HTTP → IF → Slack.
- Show a healthy run and an intentionally failed health check.
- Show the resulting Slack alert in the real environment.
- Show GitHub repository structure and documentation.
- Show execution retention configuration and conclude with production readiness.

14. Final Notes

This package intentionally distinguishes between what was implemented/verified in n8n and what requires the operator's production environment. Real external email, database writes, Slack delivery, server-level pruning, screenshots, and video evidence should only be marked complete after they are actually performed and captured.

Prepared as the documentation deliverable for Topic 12 — Production Deployment, Monitoring, and Version Control.